CTB2410 Waterbouwkunde

(Week 2)

Dr. ir. Davide Wüthrich

Assistant Professor in Hydraulic Structures

TU Delft





Davide Wüthrich

d.wuthrich@tudelft.nl



B.Sc.
Civil Engineering







2010 | 2012 | 2013 | 2014-2018 | 2018 | 2019-2020 | 2021-

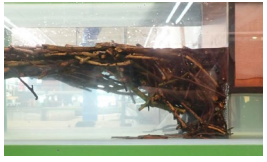
M.Sc.
Hydraulic Engineering











TU Delft

Constructies met een waterkerende functie

Structures with a water-retaining function

HYDRAULIC ~~Structures~~ SCHEMES



Water retention (Dams)



Spillways



Acqueducts



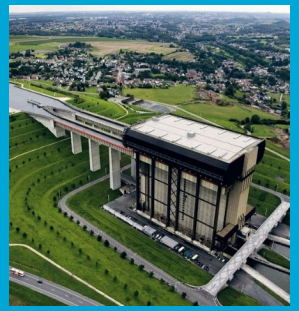
Conveyance



Flood protection



Ports and Harbours



Navigation



Barriers & Weirs



Protection



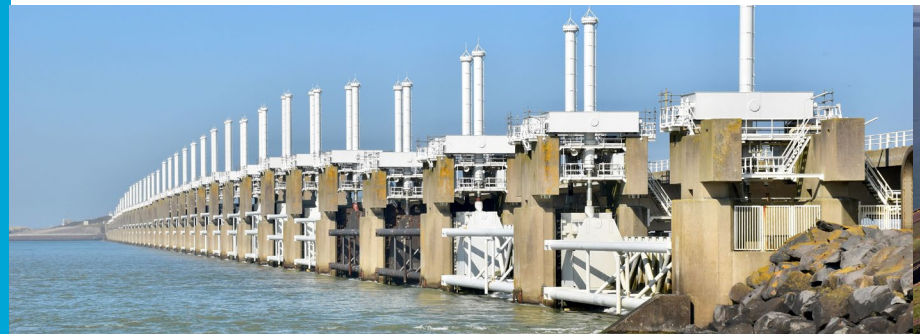
Energy production

CTB2410
Waterbouwkunde

Week 2 - Lecture 1
(D.Wüthrich)

Inhoud / *Content*

1. Stabiliteit / *Stability*
2. Analogie (ingeklemde ligger) / *Analogy (clamped beam)*
3. Methodiek / *Methodology (how do we assess stability?)*
4. Evenwicht van de constructive / *Equilibrium of the structure*
5. Bezwijkmechanismen / *Failure mechanisms*
6. Bijzondere omstandigheden / *Special circumstances*





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CTB2410
Waterbouwkunde

Week 2 - Lecture 1
(D.Wüthrich)

Stabiliteit van een waterkerende constructie

Stability of a water-retaining structure






What is it about?

1. Principe van stabiliteit / *Principle of stability*
2. Bezwijkmechanismen / *Failure mechanisms*

What is it about?

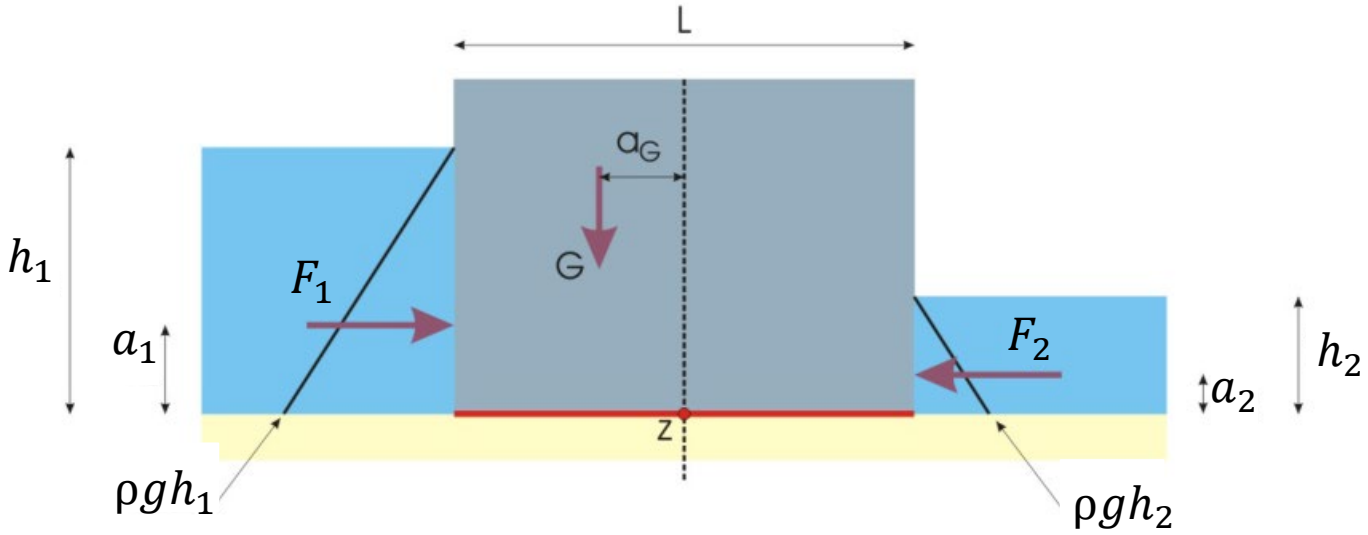






Water retention is often combined with '*another*' function.

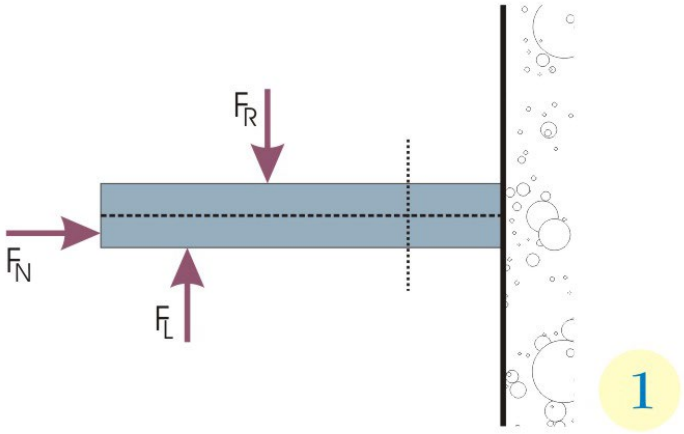
What is it about?



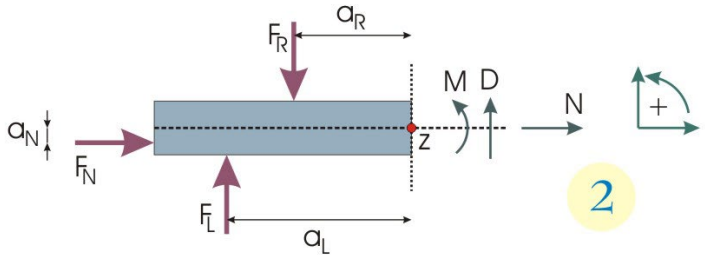
Loads are maintained **stationary** : water levels on either side of the structure do not change over time (no tides, no waves yet).

Unilaterally restrained, prismatic, homogeneous structures on which several (**deterministic**) point loads act. Probabilistic approaches (mentioned in week 1) won't be used in week 2.

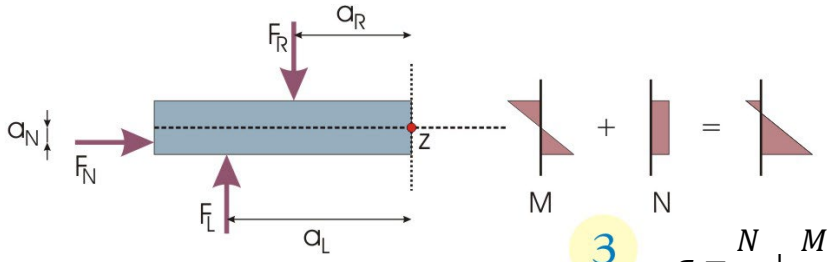
Analogie / *Analogy*



1



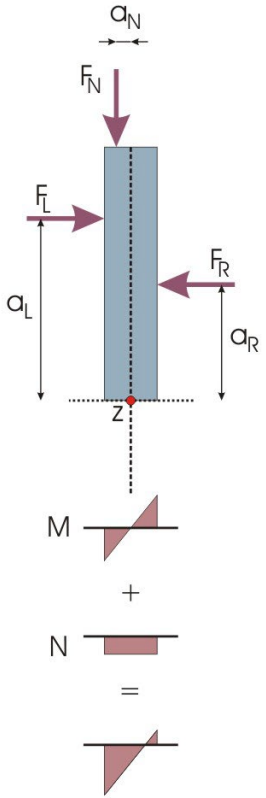
2

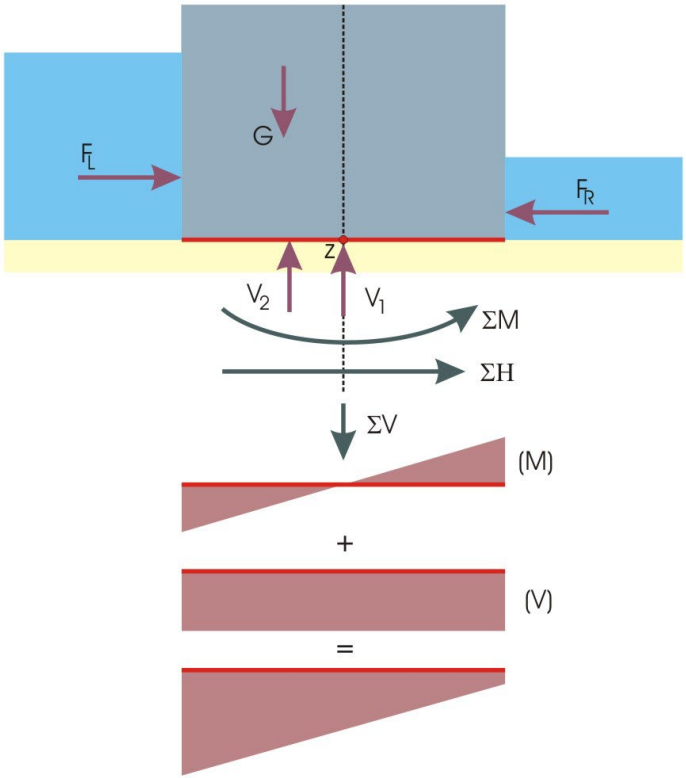


3

$$\sigma = \frac{N}{A} \pm \frac{M}{W}$$

Analogie / *Analogy*





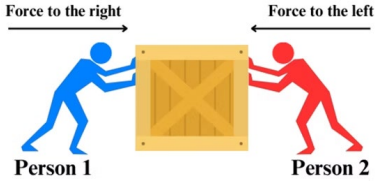
Methodiek / *Methodology*

Inventarisatie van belastingen / *Inventory of loads*

1. Verloop van de waterdruk rond de constructive /
   Water pressure variation around the structure
2. Eigen gewicht / *Self-weight*
3. Overige (externe) belastingen / *Other (external) loads*

Oplegreactie vanuit fundering op basis van horizontaal en vertikaal evenwicht en evenwicht van momenten
Support from the foundation is based on horizontal and vertical equilibrium and equilibrium of moments.

Volstaat de fundering? / *Is the foundation sufficient?*

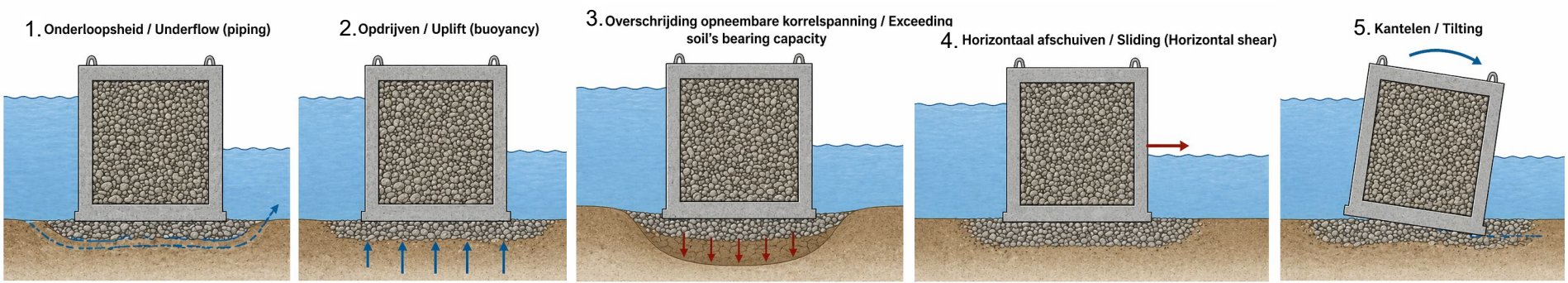


<u>Equilibrium can only be achieved if these stresses can be supplied by the subsoil!</u>

Methodiek / *Methodology*

Volstaat de fundering? → Bezwijkmechanismen
Is the foundation sufficient? → *Failure mechanisms*

1. Onderloopsheid / *Underflow (piping)*
2. Opdrijven / *Uplift (buoyancy)*
3. Overschrijding opneembare korrelspanning / *Exceeding soil's bearing capacity*
4. Horizontaal afschuiven / *Sliding (Horizontal shear)*
5. Kantelen / *Tilting*



Methodiek / *Methodology*

In dit vak / *In this subject:*

1. Elementaire (op staal gefundeerde) constructive / *Elementary (gravity-based) structures*
   - Inventarisatie van belastingen / *Inventory of loads (water pressure profile)*
   - Evenwicht (oplegreactie) / *Equilibrium*

2. Bezwijkmechanismen / *Failure mechanisms (consequences and measures)*
   - Onderloopsheid / *Underflow*
   - Opdrijven / *Uplift (Buoyancy)*
   - Afschuiven / *Sliding*
   - Kantelen / *Tilting*

3. Bijzondere omstandigheden / *Special circumstances*
   - Paalfundering / *Pile foundation*
   - Water in de constructive / *Water in (or through) the structure*

Methodiek / *Methodology*

Voor berekeningen hanteren we / *For calculations, we use*:

- naar rechts / *to the right*
- naar boven / *upwards*
- linksom / *counterclockwise*

positief

- naar links / *to the left*
- naar beneden / *downwards*
- rechtsom / *clockwise*

negatief



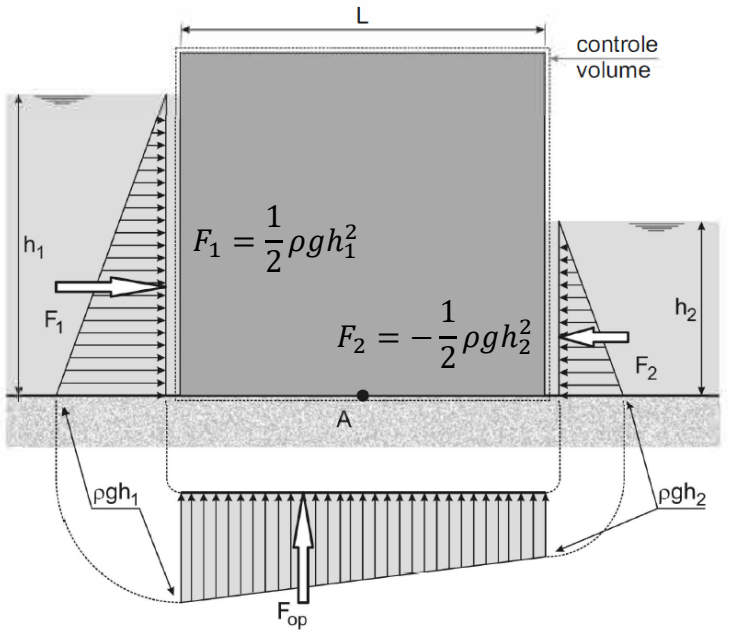
Andere keuze is ook goed, mits daar consequent mee wordt omgesprongen.
Another choice is also fine, provided it is handled consistently.

Vermeld altijd uw keuze zodat anderen begrijpen wat u gedaan heeft!
Always state your choice so that others understand what you have done!

Evenwicht / *Equilibrium*

Verloop van de waterdruk / *Hydrostatic pressure*



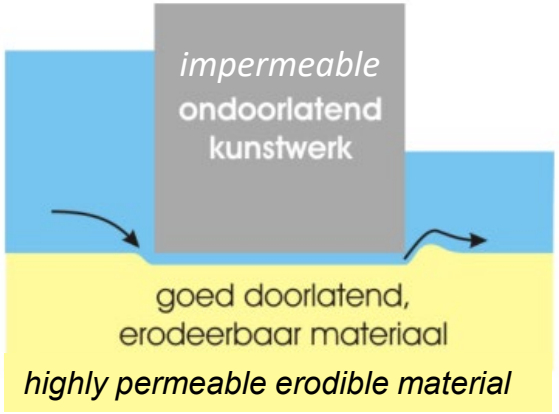


Aanname: stijghoogte verloopt lineair langs het grensvlak tussen constructie en ondergrond.
Assumption: piezometric head varies linearly at the interface between structure and subsoil.

⚠ Gradient → Possible (under)flow

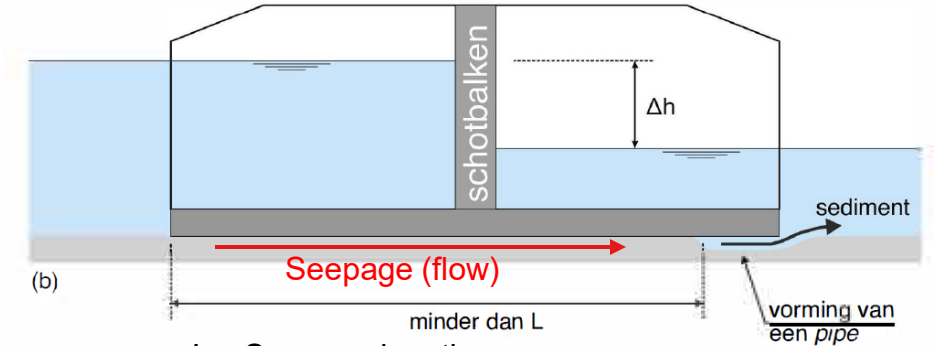
Onderloopsheid / *Underflow* (piping)

Mechanism:



The water flowing along or under the structure is called '*seepage*'

Seepage can carry sediment, creating small channels next to or under the structure with preferential flows. Such a channel is called a '***pipe***'.



L_k : Seepage length

$$u = -k_p \frac{\Delta h}{L_k}$$

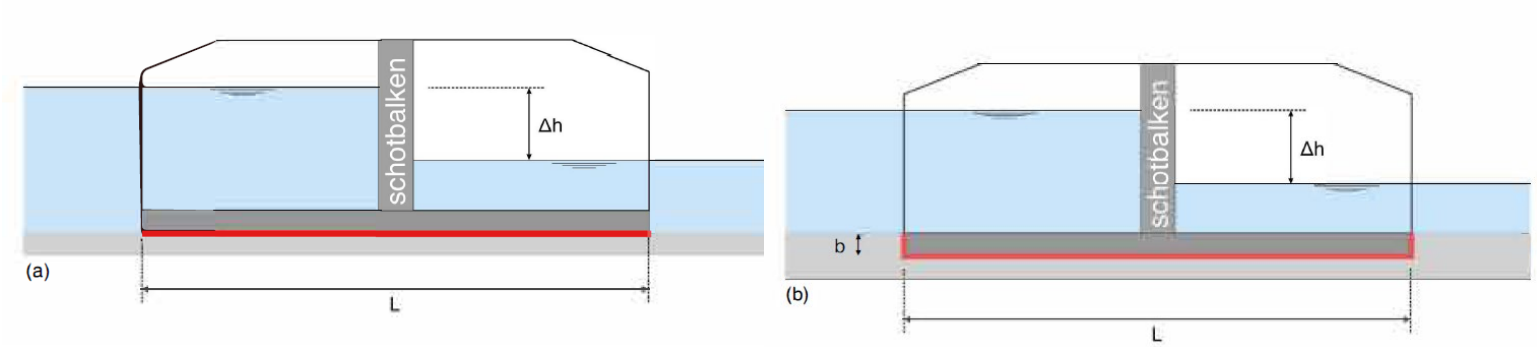
permeability and porosity of the subsurface

$$L_k > \frac{k_p}{|u_{crit}|} \Delta h$$

Onderloopsheid / *Underflow* (piping)

$$L_{\mathrm{k}} = L_{\mathrm{k,hor}} + a \cdot L_{\mathrm{k,vert}} > 1.5 \cdot C \cdot \Delta h$$

met Δh het verval over de constructie en C een coefficient (afhankelijk van de ondergrond).
with Δh the head loss across the structure and C a coefficient (dependent on the subsoil).



Methode Bligh (1910):
$C = C_B$ (empirisch / *empirical* ; getabelleerd / *tabulated*)
$a = 1$

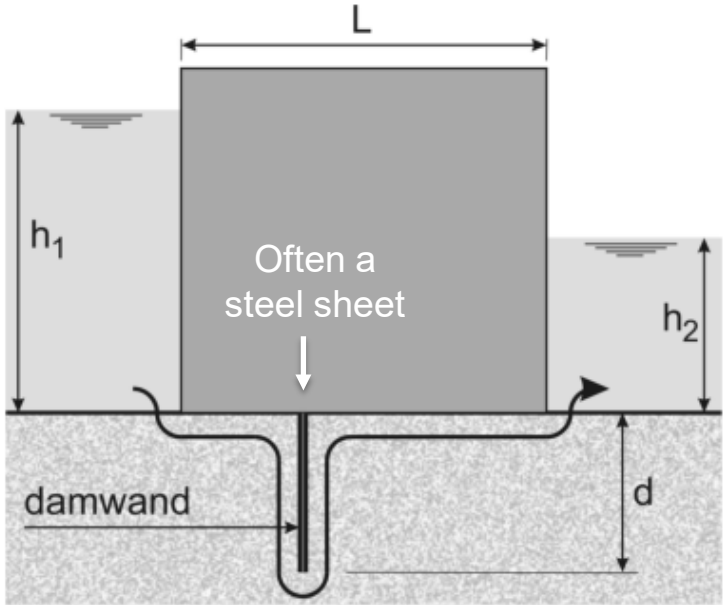
Methode Lane (1935):
$C = C_L$ (empirisch / *empirical* ; getabelleerd / *tabulated*)
$a = 3$

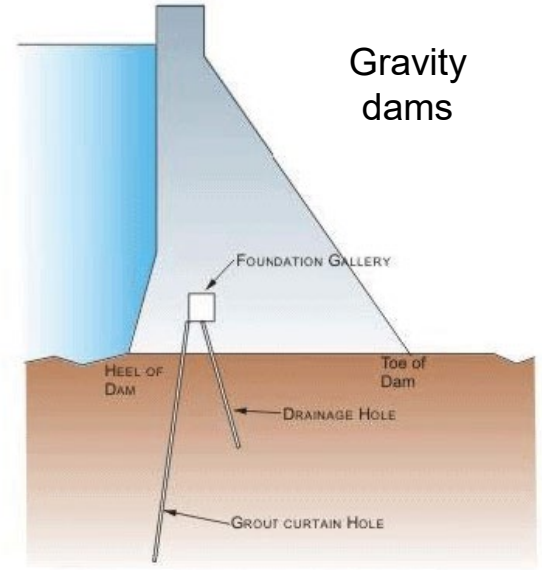
Onderloopsheid / *Underflow* (piping)

$$L_{\mathrm{k}} = L_{\mathrm{k,hor}} + a \cdot L_{\mathrm{k,vert}} > 1.5 \cdot C \cdot \Delta h$$

How can I maximize the seepage length?

1. *Extend the width of the structure (not always possible and/or economical)*
2. *Add a vertical seepage screen (*Onderloopsheidscherm*)*





Constructies met een waterkerende functie:

1. *General Stability*
2. ***Underflow***
3. *Uplift*
4. *Soil capacity*
5. *Sliding*
6. *Tilting*
7. *Pile foundation*
8. *Flow in the struct.*

Questions?

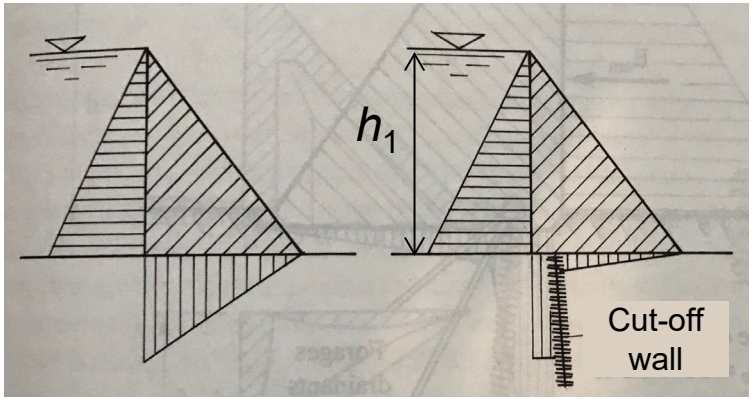


Opdrijven / *Uplift*

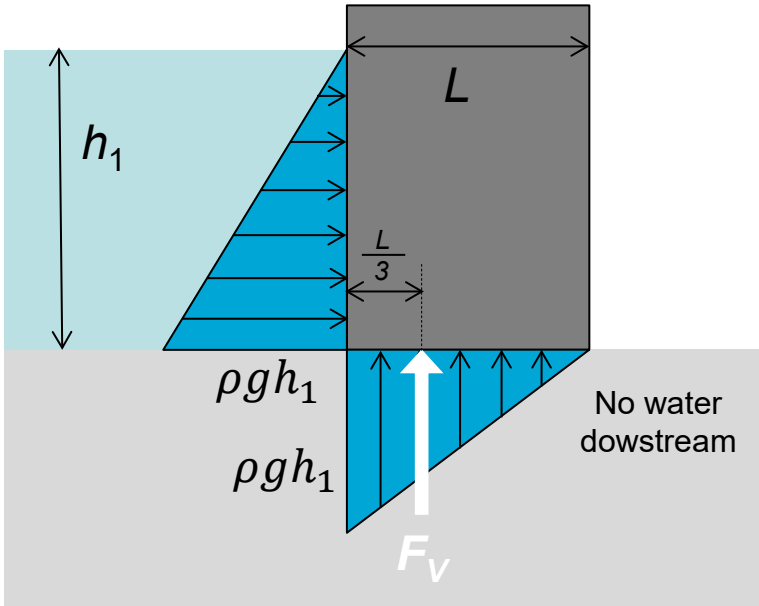
The water enters the pores, fissures, and cracks of the foundation

The seeping water creates hydraulic gradient between the upstream and downstream side of the structure ⟶ **vertical upward pressure**

$$F_{\text{op}} = \frac{1}{2}\rho g h_1 L B$$



$h_{\text{cut-off}}$ = 0.3-1·h_1

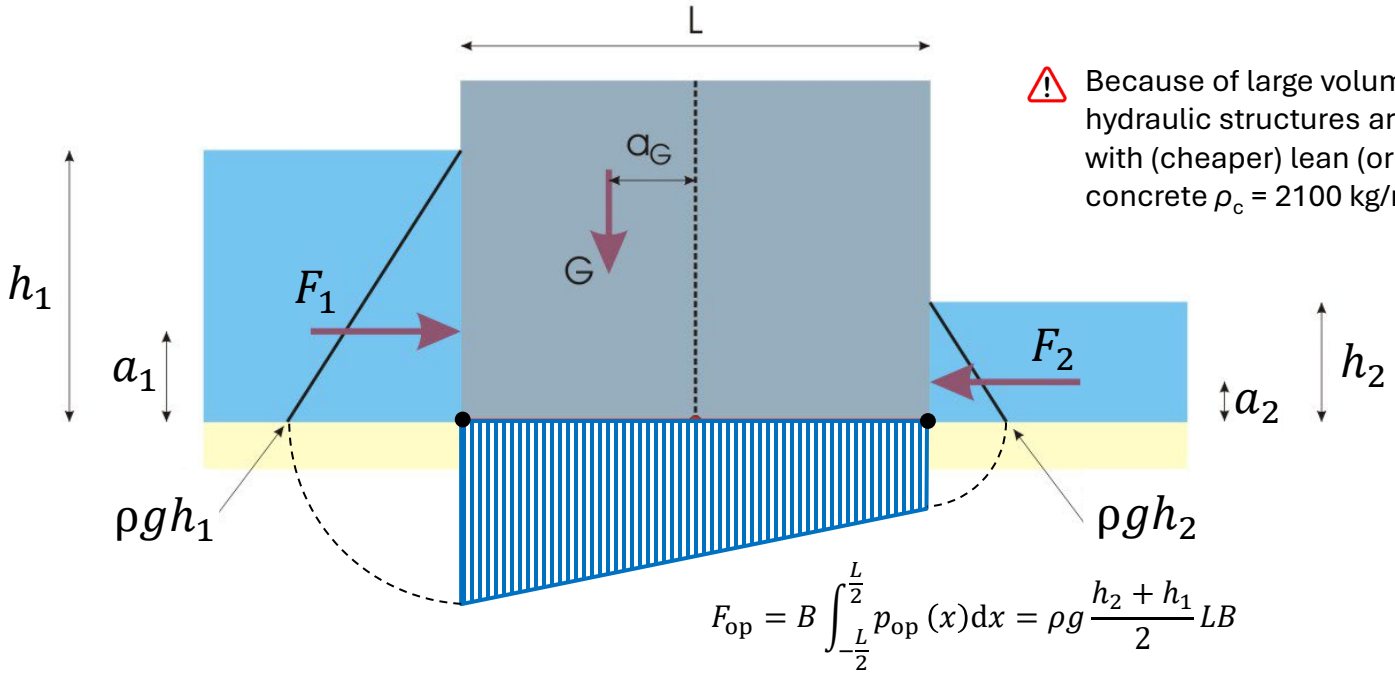


Constructies met een waterkerende functie:

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Opwaartse waterdruk / *Uplift*

Water is beneath the ground surface, which causes an upward load



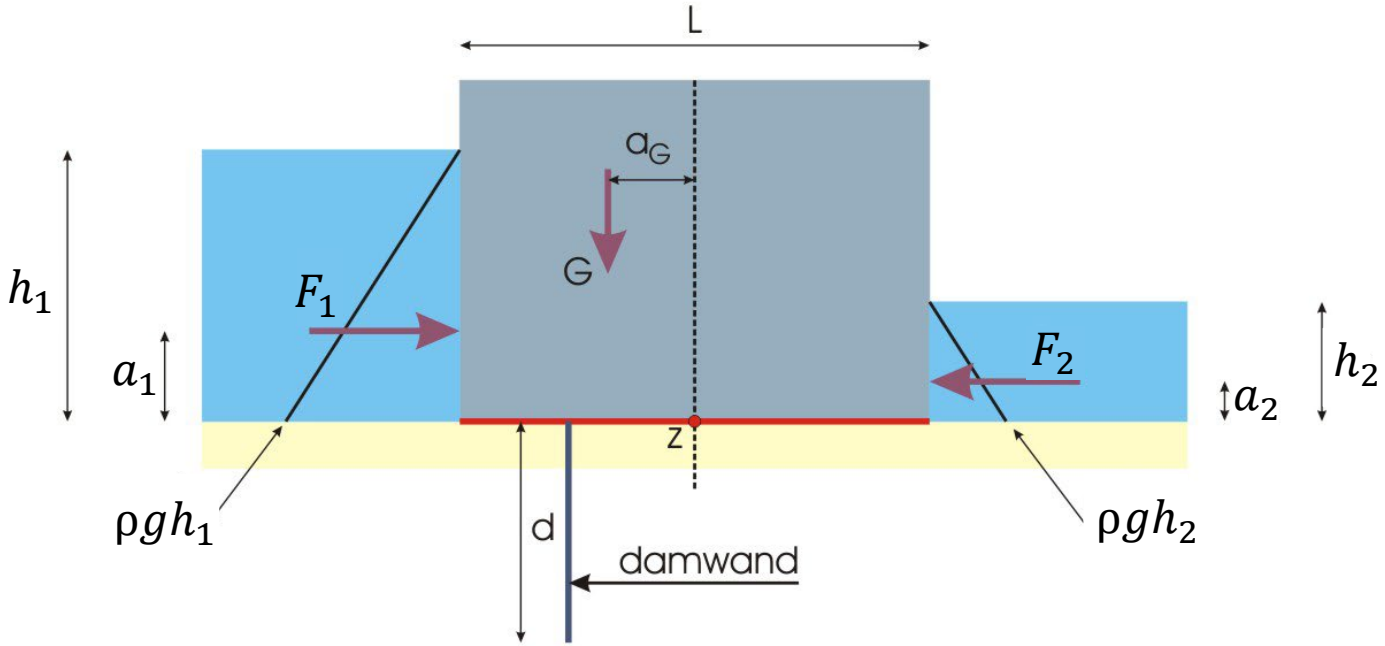
⚠ Because of large volumes hydraulic structures are built with (cheaper) lean (or mass) concrete ρ_c = 2100 kg/m^3

$$F_{op} = B\int_{-\frac{L}{2}}^{\frac{L}{2}} p_{op}(x)\mathrm{d}x = \rho g \frac{h_2 + h_1}{2} LB$$

To ensure (vertical) stability → *G* (weight of the structure) > F_{op}

Opwaartse waterdruk / *Uplift*

A seepage wall is now added to prevent piping

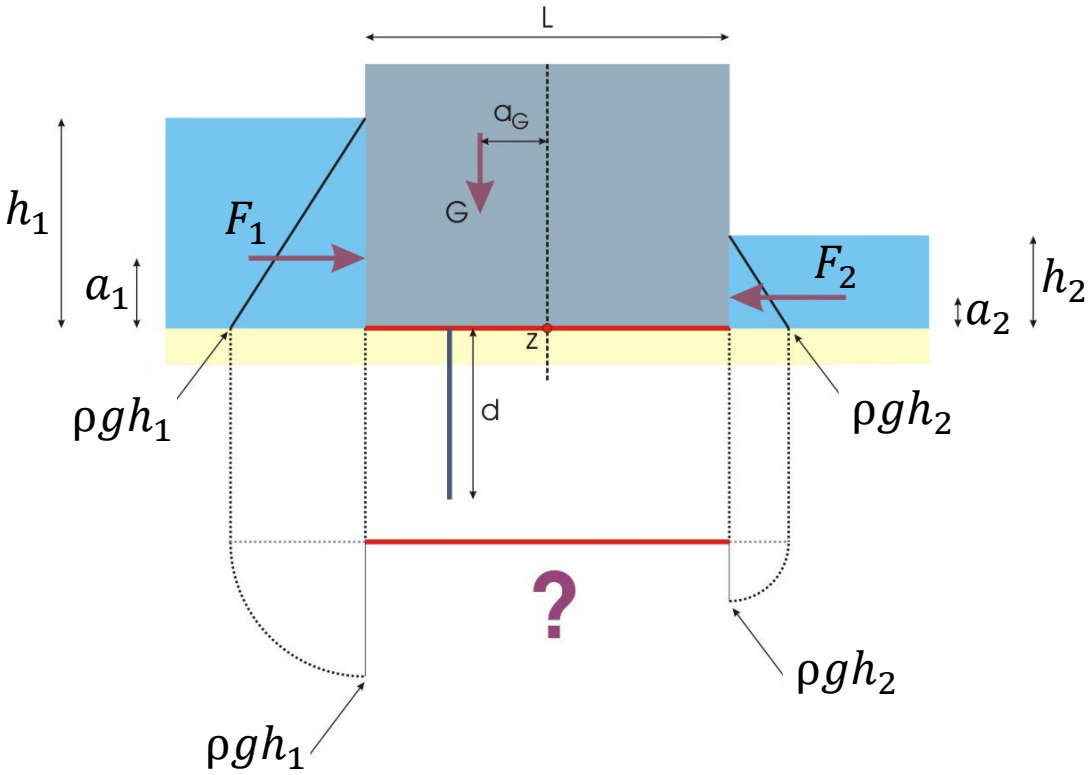


How does it affect the under-pressures?

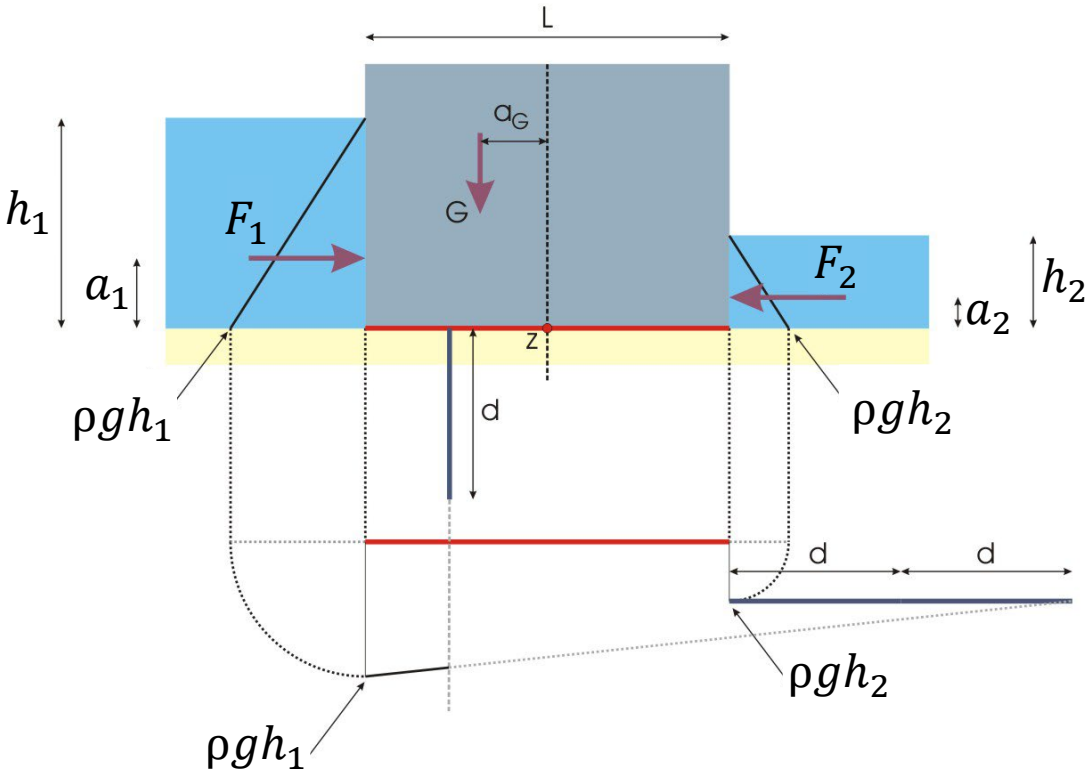
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Opwaartse waterdruk / *Uplift*



Opwaartse waterdruk / *Uplift*

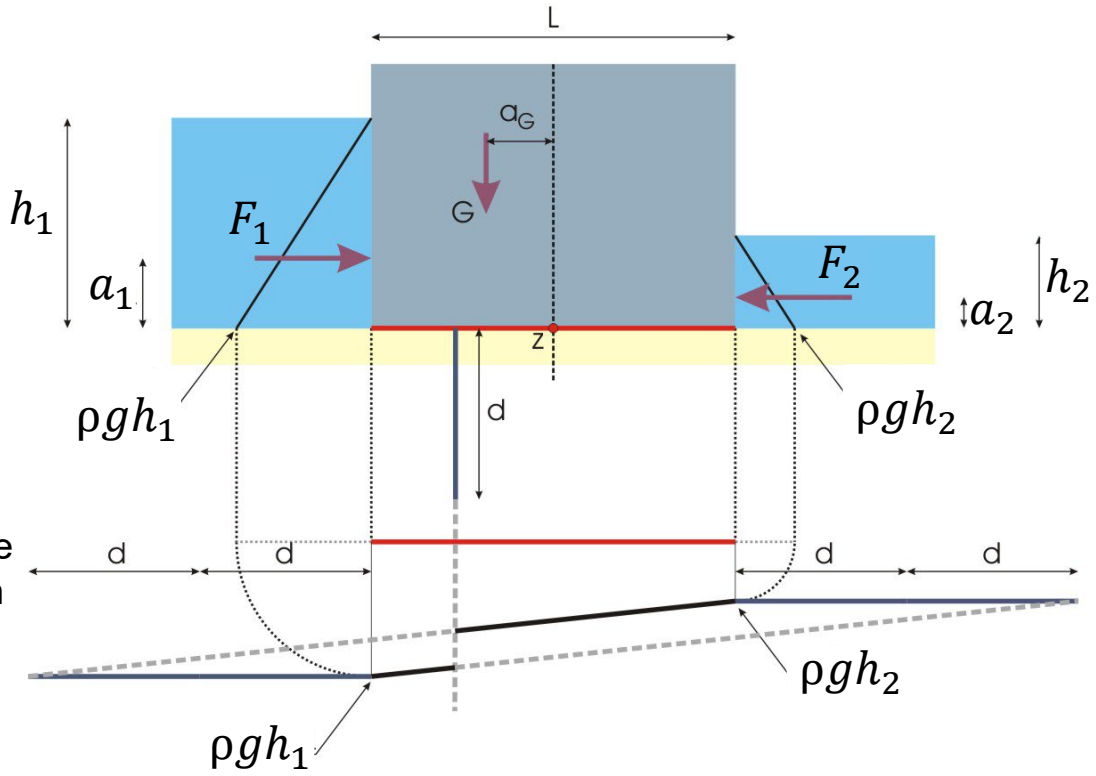


Add 2 × *d* to the seepage length

Constructies met een waterkerende functie:

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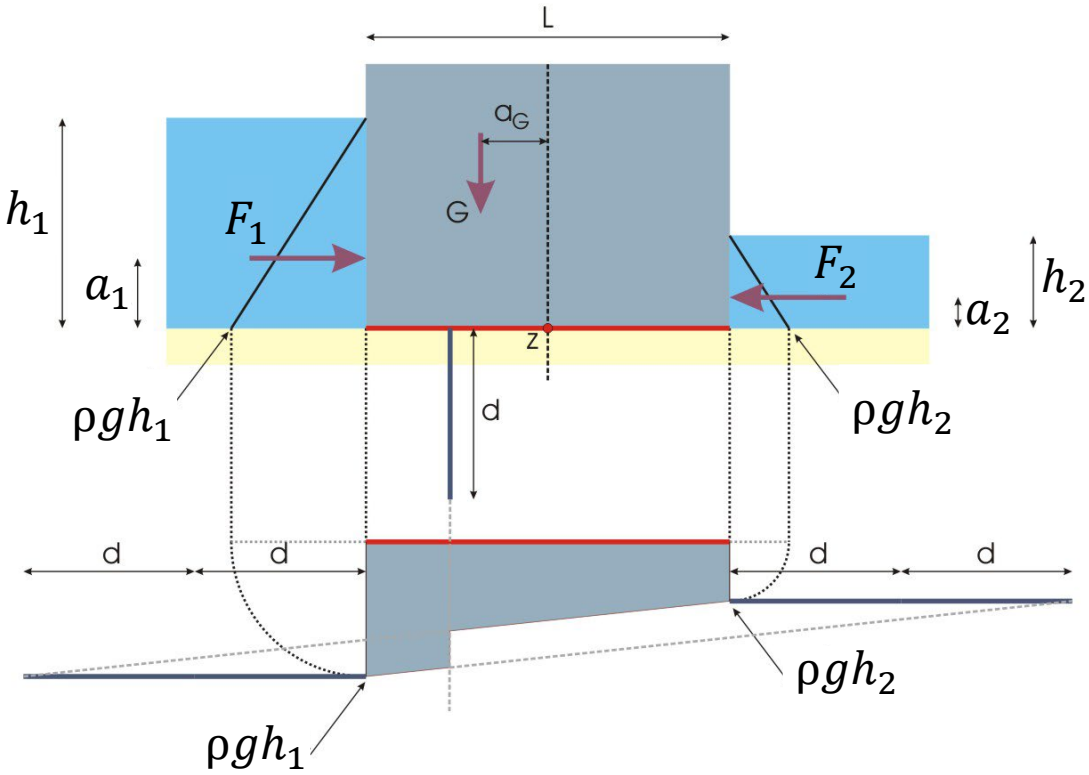
Opwaartse waterdruk / *Uplift*



Add 2 × *d* to the seepage length

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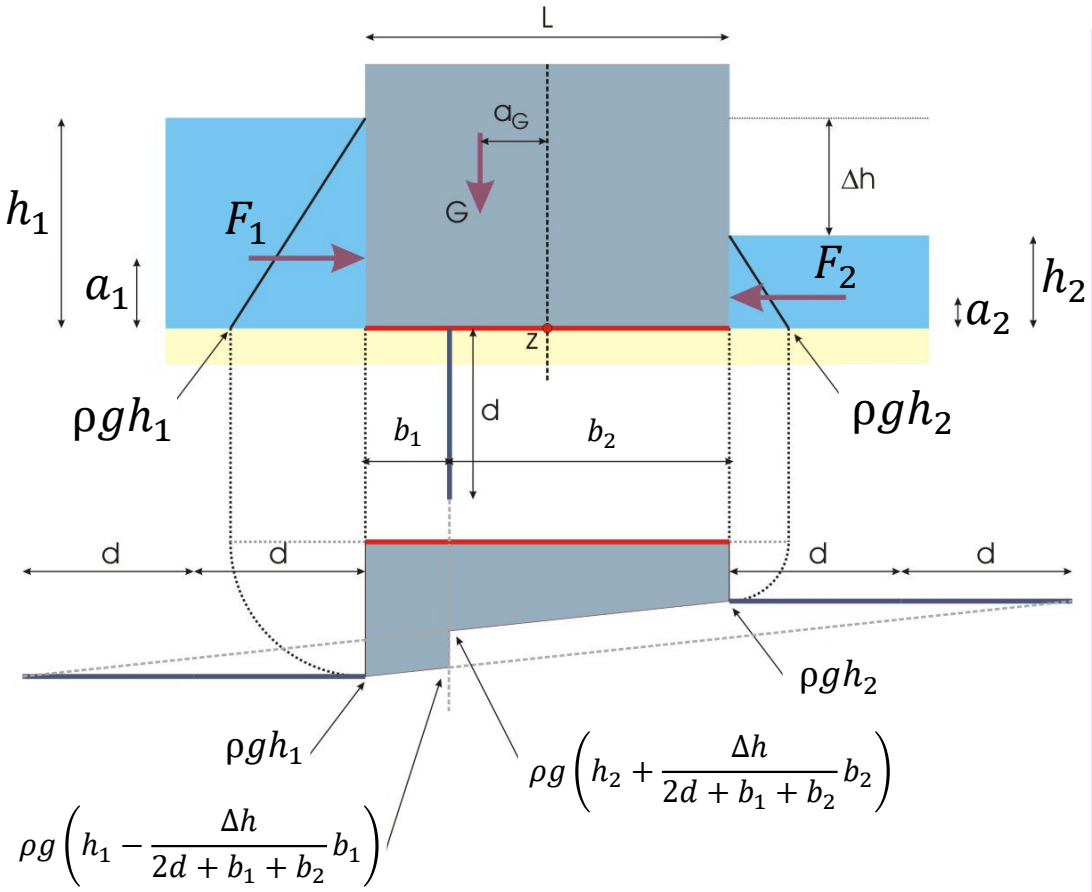
Opwaartse waterdruk / *Uplift*



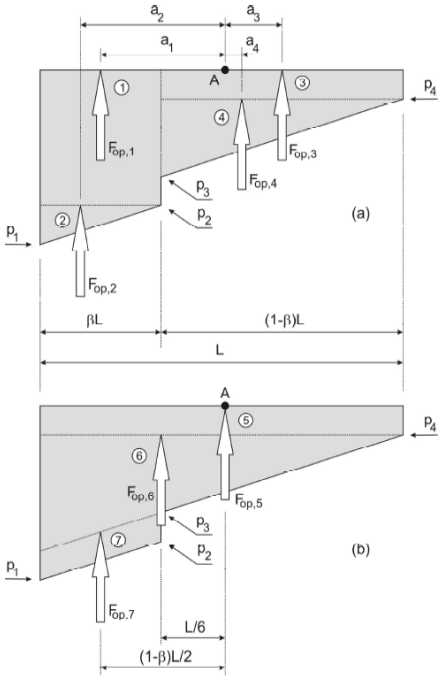
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Opwaartse waterdruk / *Uplift*



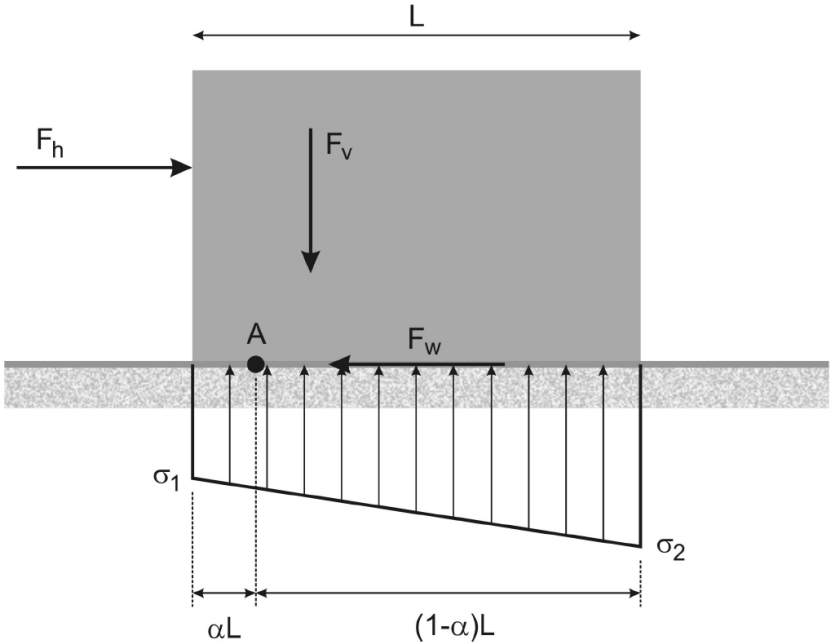
Resultant & Moments



Questions?



Evenwicht / *Equilibrium* (soil capacity)



Oplegreactie vanuit de ondergrond
Support reaction from the subsurface

Horizontaal:

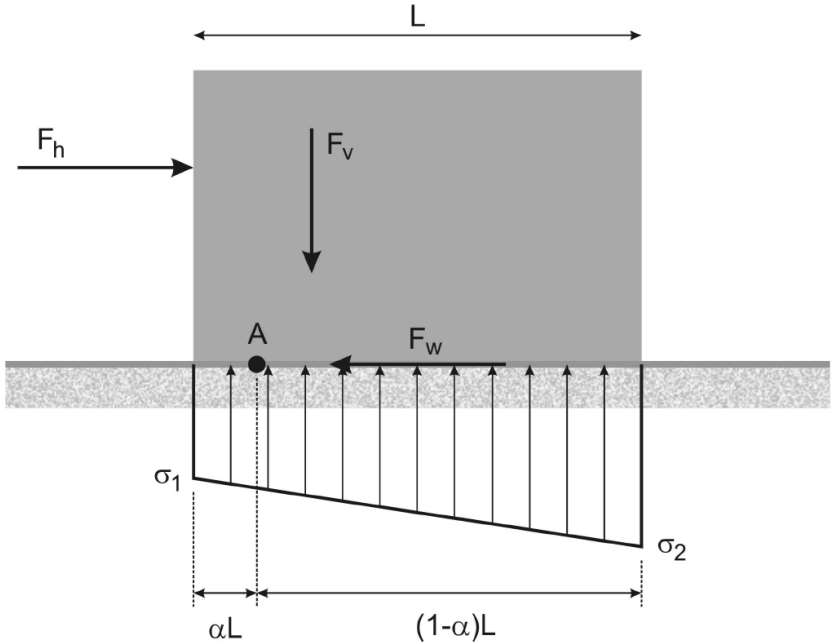
$$F_{\mathrm{h}} + F_{\mathrm{w}} = 0$$

Vertikaal:

$$F_{\mathrm{v}} + \frac{\sigma_1 + \sigma_2}{2} \cdot B \cdot L = 0$$

Aanname: Ondergrond is uniform en gedraagt zich lineair elastisch
Assumption: The (soil) substrate is uniform and behaves linearly elastically. Drained conditions in case of soil deformations.

Evenwicht / *Equilibrium* (soil capacity)



Momenten / *Moments*:

$$M_h + M_v(\alpha) + M_k = 0$$

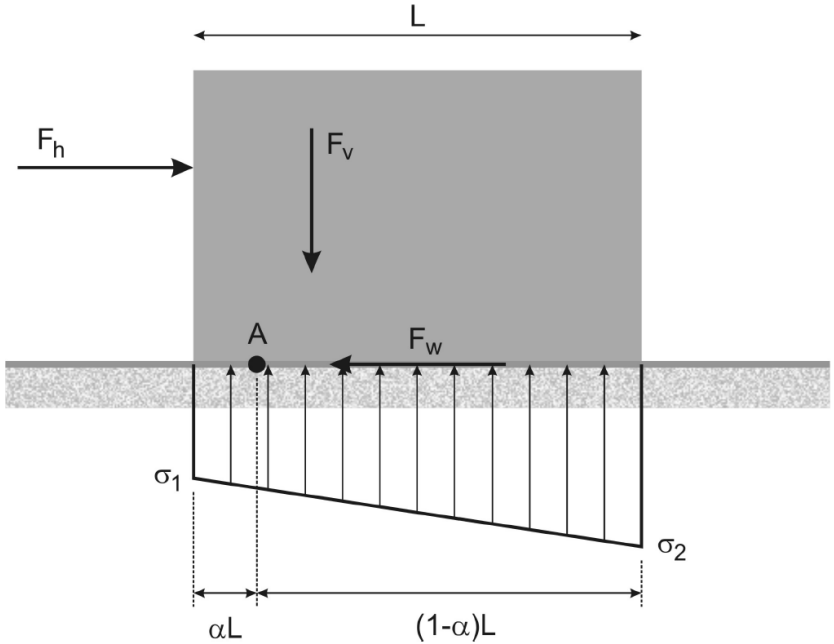
- M_h : independent of α
- Friction: no moment at A

$$M_k = BL^2 \frac{(1-3\alpha)\sigma_1 + (2-3\alpha)\sigma_2}{6}$$

or with $W = \frac{BL^2}{6}$ it becomes:

$$\frac{M_k}{W} = (1-3\alpha)\sigma_1 + (2-3\alpha)\sigma_2$$

Evenwicht / *Equilibrium* (soil capacity)



We can solve for σ_1 and σ_2

$$\sigma_1 = (6\alpha - 4)\frac{F_v}{BL} + \frac{M_h + M_v(\alpha)}{W}$$

$$\sigma_2 = (2 - 6\alpha)\frac{F_v}{BL} - \frac{M_h + M_v(\alpha)}{W}$$

Handige keuze / *Convenient choice*

α = 0.5 (center)

$$\sigma_1 = -\frac{F_v}{BL} + \frac{M_h + M_v(1/2)}{W}$$

$$\sigma_2 = -\frac{F_v}{BL} - \frac{M_h + M_v(1/2)}{W}$$

$W = \frac{BL^2}{6}$ weerstandsmoment van het grondvlak
section modulus of the base

Constructies met een waterkerende functie:

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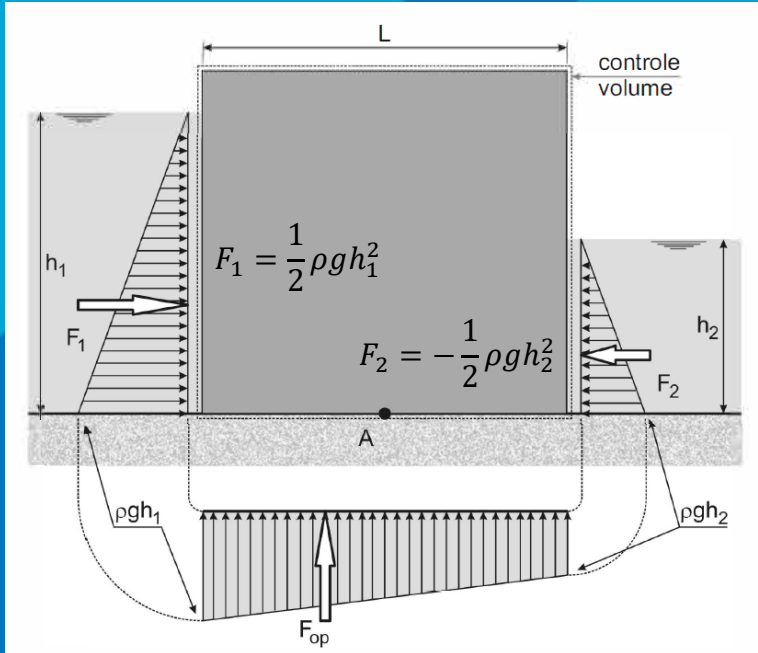
Questions?



CTB2410 Waterbouwkunde
(Week 2)

Constructies met een waterkerende functie:

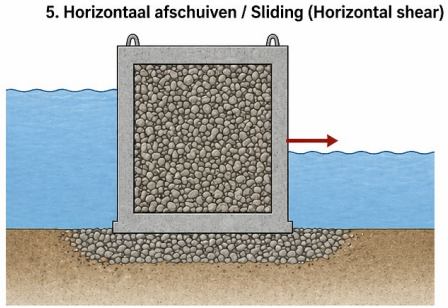
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Afschuiven / *Sliding*



Wrijving / *Friction*

$$|\sigma_w| \leq f_w |\sigma_v|$$

Met *A* het oppervlak van het grondvlak / *With A the area of the base*

Maximale wrijvingskracht / *Maximum friction force*

$$|F_{w,\max}| = f_w |F_v|$$

Interface Materials	Friction Coefficient, f
Mass concrete on the following foundation materials:	
Clean sound rock	0.70
Clean gravel, gravel--sand mixtures, coarse sand	0.55 to 0.60
Clean fine to medium sand, silty medium to coarse sand, silty or clayey gravel	0.45 to 0.55
Clean fine sand, silty or clayey fine to medium sand	0.35 to 0.45
Fine sandy silt, nonplastic silt	0.30 to 0.35
Very stiff and hard residual or preconsolidated clay	0.40 to 0.50
Medium stiff and stiff clay and silty clay	0.30 to 0.35

Friction coefficient for cast concrete on soil [USACE Technical Letters]

Afschuiven / *Sliding*

For design: maximize f_w ⟶ There is a limit : $\tan\delta$ (function of soil properties)

$$|F_{w,\max}| = \min\{f_w; \tan\delta\}|F_v|$$

Sterkte & belasting / *Strength & load*:

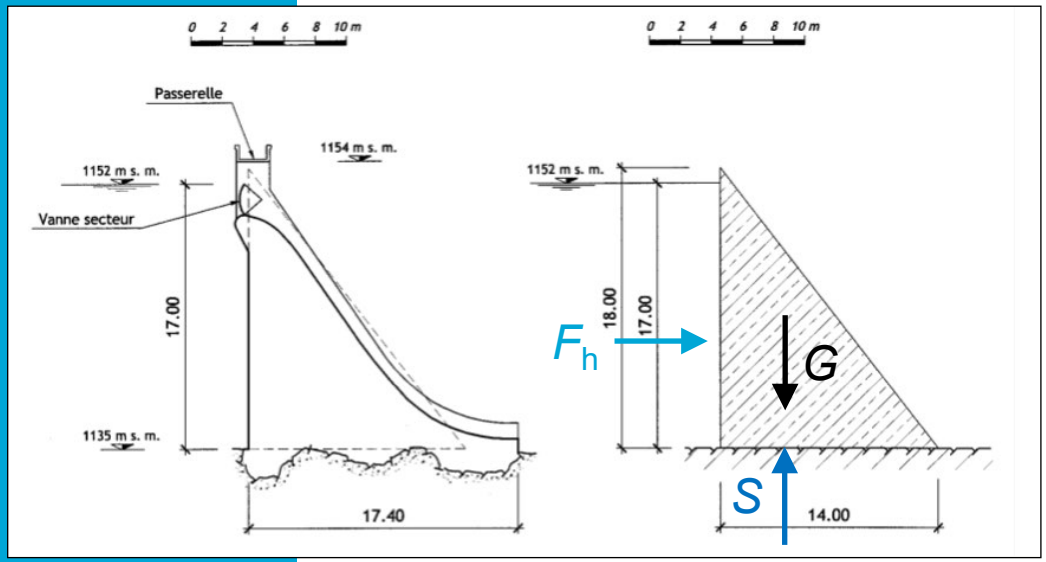
$$\gamma = \frac{|F_{w,\max}|}{|F_h|} = \frac{\min\{f_w; \tan\delta\}|F_v|}{|F_h|}$$

Eis / *Requirement* $\gamma > 1$

Note: The angle ϕ of internal friction of the soil is not always well known in practice. To address this uncertainty, an angle δ is generally used that is $2/3\ \phi < \delta < \phi$. This a **safety choice** made by the designer.







$$F_h = \frac{1}{2}\rho_w g h_w^2 = 1417 \text{ kN/m}$$

$$S = \frac{1}{2}\rho_w g h_w L = -1167 \text{ kN/m}$$

$$G = \frac{1}{2}\rho_c g h_{\text{dam}} L = 3090 \text{ kN/m}$$

Assuming (e.g.) a density of concrete ρ_c = 2500 m^3/s and f_w = 0.7 for sound rock (previous slide).

$$\gamma = \frac{(G - S) f_w}{F_h} = 0{,}948$$

IT WILL SLIDE

Constructies met een waterkerende functie:

1. *General Stability*
2. *Underflow*
3. *Uplift*
4. *Soil capacity*
5. ***Sliding***
6. *Tilting*
7. *Pile foundation*
8. *Flow in the struct.*

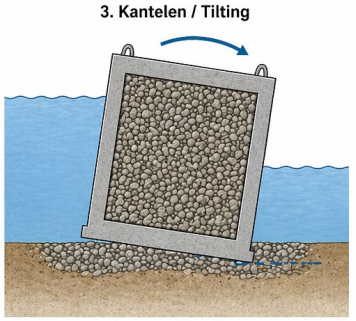
Questions?



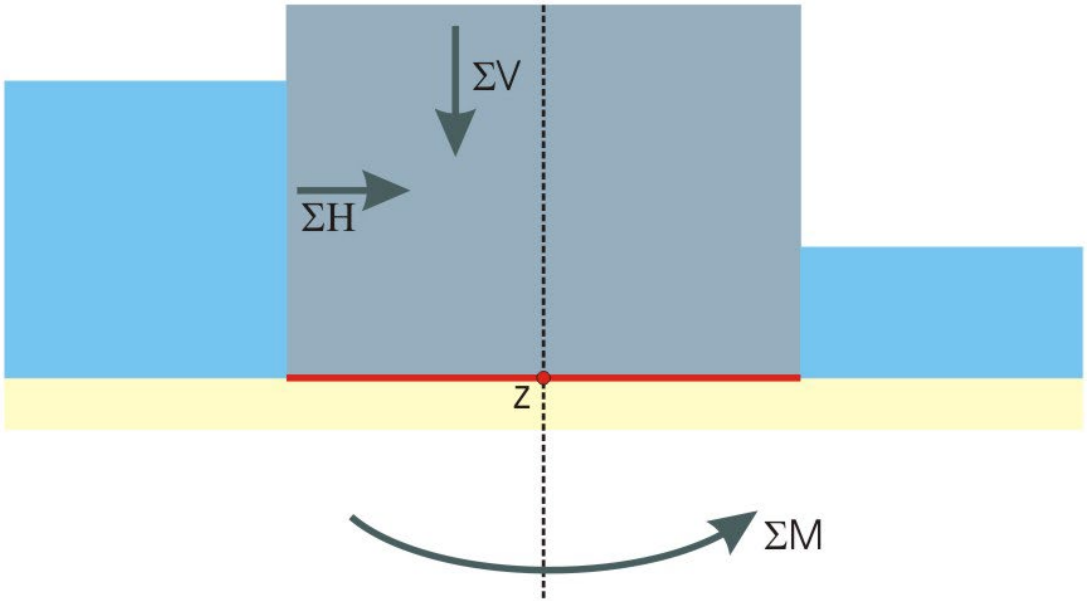
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Kantelen / *Tilting*

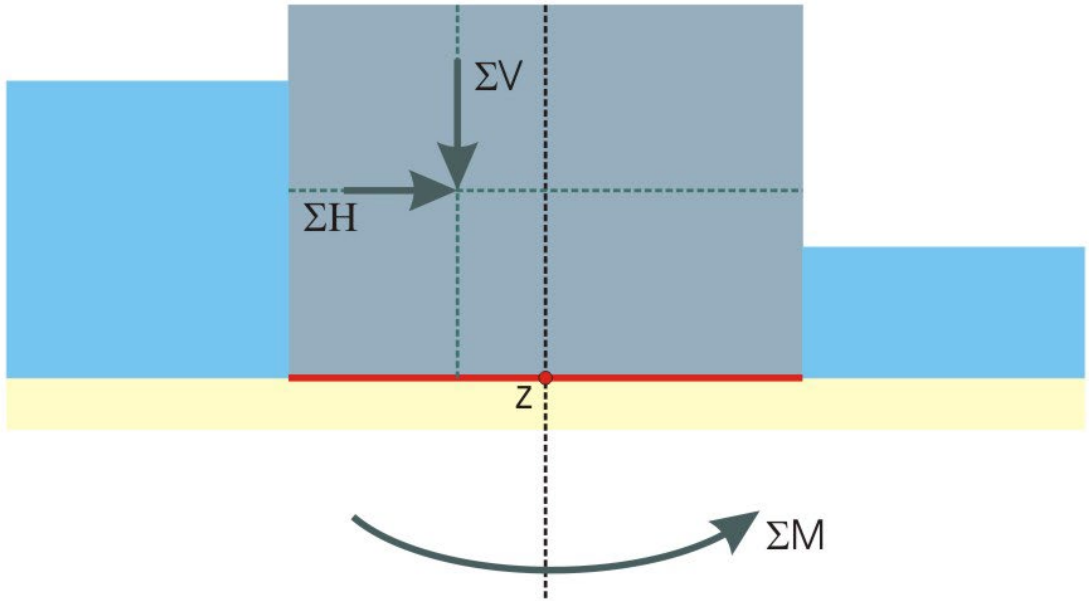


All the loads ⟶ $F_v = \Sigma V$ $F_h = \Sigma H$



Kantelen / *Tilting*

All the loads ⟶ $F_v = \Sigma V$ $F_h = \Sigma H$

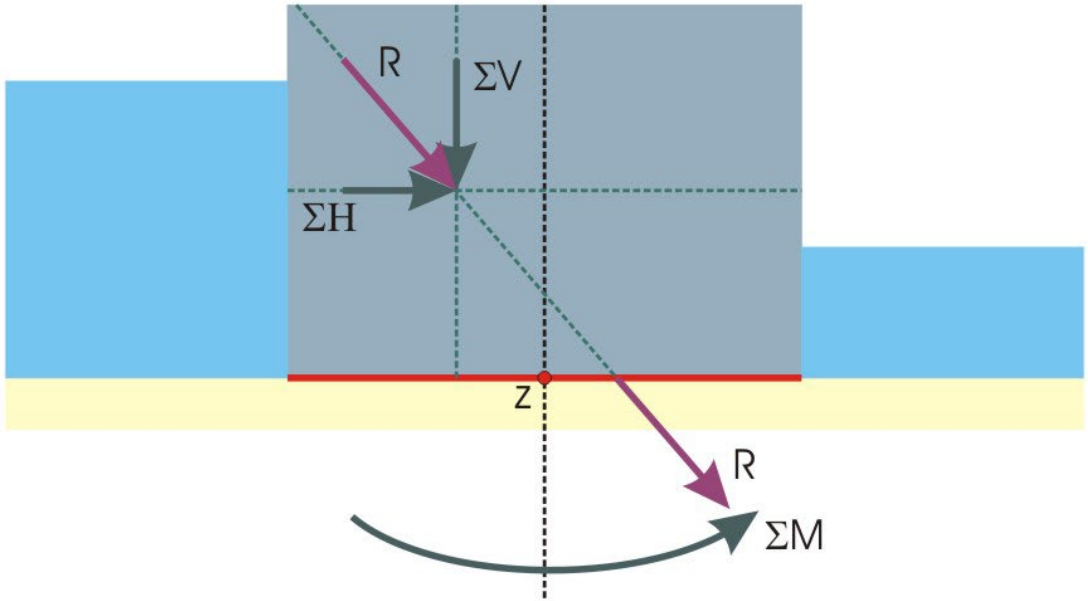


Constructies met een waterkerende functie:

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Kantelen / *Tilting*

All the loads → $F_v = \Sigma V$ $F_h = \Sigma H$ → Resultant **R**

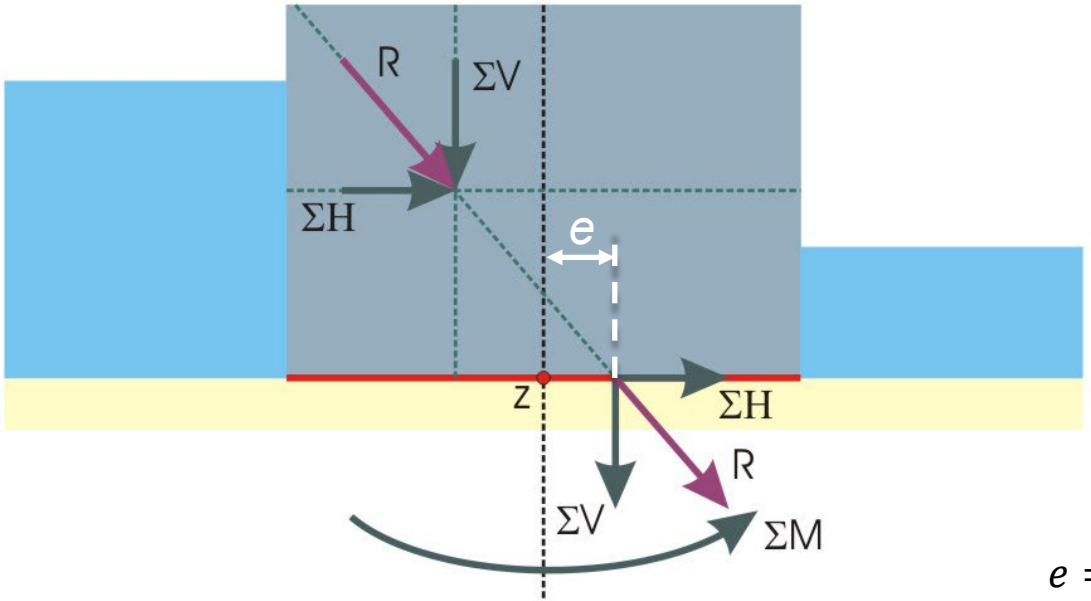


Constructies met een waterkerende functie:

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Kantelen / *Tilting*

All the loads → $F_v = \Sigma V$ $F_h = \Sigma H$ → Resultant ***R***



Vertical and Horizontal Components ΣV and ΣH

Focus on ΣV (or F_v)

Eccentricity *e* (Excentriciteit)

$$e = \frac{\Sigma M}{F_v}$$

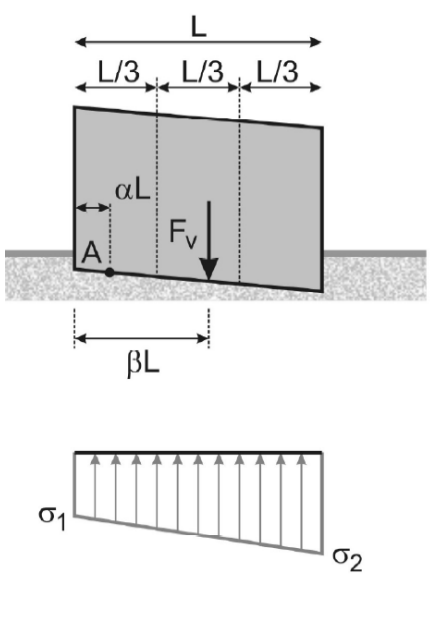
Constructies met een waterkerende functie:

1. *General Stability*
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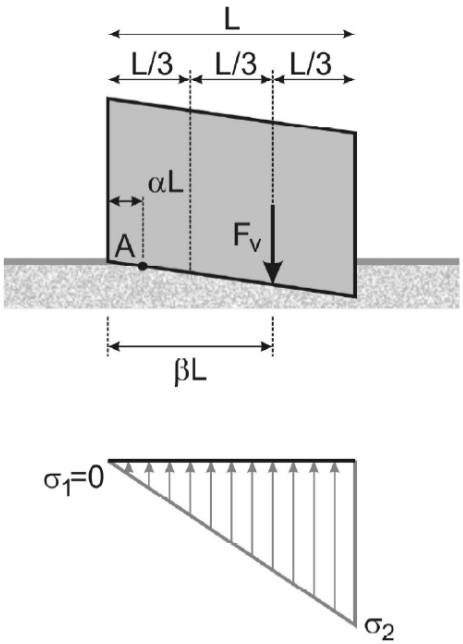
Kantelen / *Tilting*

F_v outside of the central tier

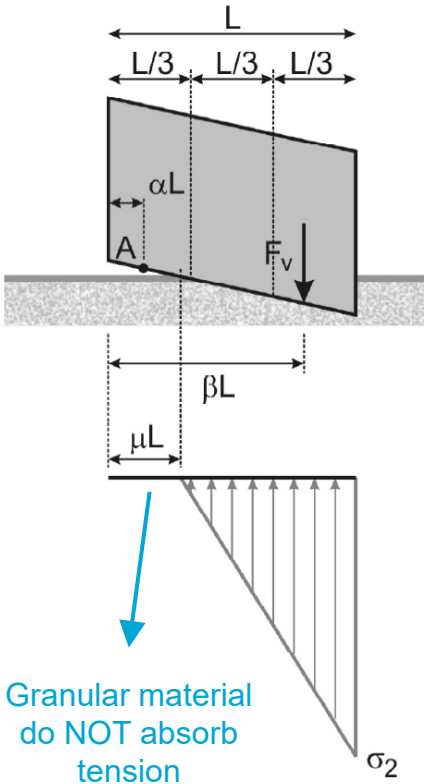
CASE 1 : $e < \frac{L}{6}$



CASE 2 : $e = \frac{L}{6}$



CASE 3 : $e > \frac{L}{6}$



As F_v shifts to the right , stress σ_2 increases and σ_1 decreases to zero (CASE 2)

Granular material do NOT absorb tension

Constructies met een waterkerende functie:
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Kantelen / *Tilting*

Uiterste toestanden / *Extreme states (mathematically, from slide 30)*

$$\sigma_1 = (6\alpha - 4)\frac{F_v}{BL} + \frac{M_h + M_v(\alpha)}{W} = 0$$

$$\Longrightarrow \quad \frac{M_h + M_v(\alpha)}{F_v} - \left(\frac{1}{2} - \alpha\right) L = \frac{1}{6} L$$

$$\sigma_2 = (2 - 6\alpha)\frac{F_v}{BL} - \frac{M_h + M_v(\alpha)}{W} = 0$$

$$\Longrightarrow \quad \frac{M_h + M_v(\alpha)}{F_v} - \left(\frac{1}{2} - \alpha\right) L = -\frac{1}{6} L$$

Kantelen / *Tilting*

$$-\frac{1}{6}L \leq \frac{M_h + M_v(\alpha)}{F_v} - \left(\frac{1}{2} - \alpha\right) L \leq \frac{1}{6}L$$

Met α = 0.5 wordt dit / *With α = 0.5, this becomes*

$$-\frac{1}{6}L \leq \frac{M_h + M_v(1/2)}{F_v} \leq \frac{1}{6}L$$

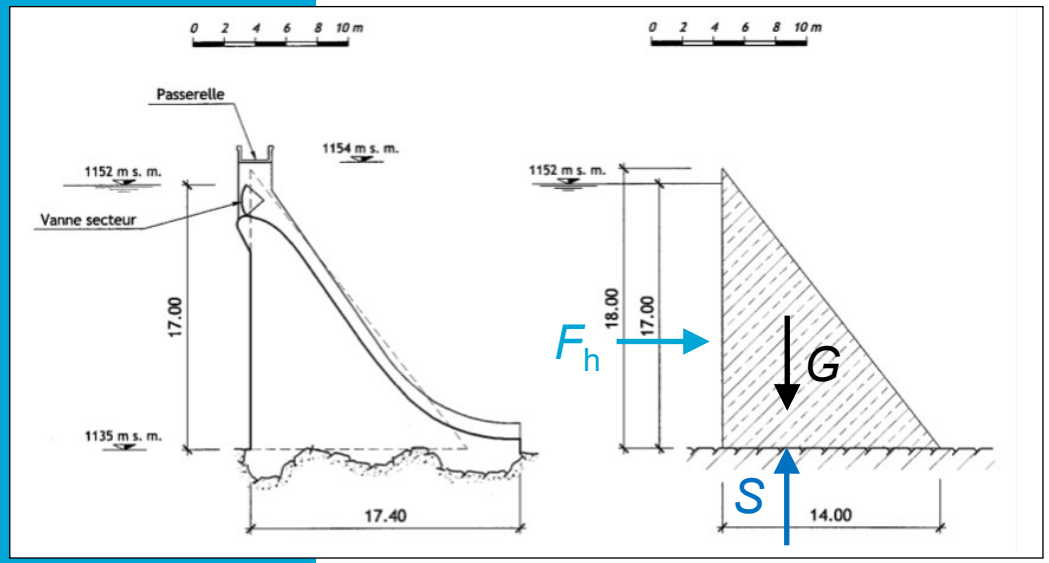
De term in het midden wordt de excentriciteit e genoemd / *The term in the middle is the eccentricity e*

$$e = \frac{M_h + M_v(\alpha)}{F_v} - \left(\frac{1}{2} - \alpha\right) L$$

$$e = \frac{M_h + M_v(1/2)}{F_v} \quad \textit{(for α = 0.5)}$$







$$M_h = F_h \cdot \frac{17\text{m}}{3} = -8032\ \text{kN/m}$$

$$M_S = S \cdot \frac{14\text{m}}{6} = -2723\ \text{kN/m}$$

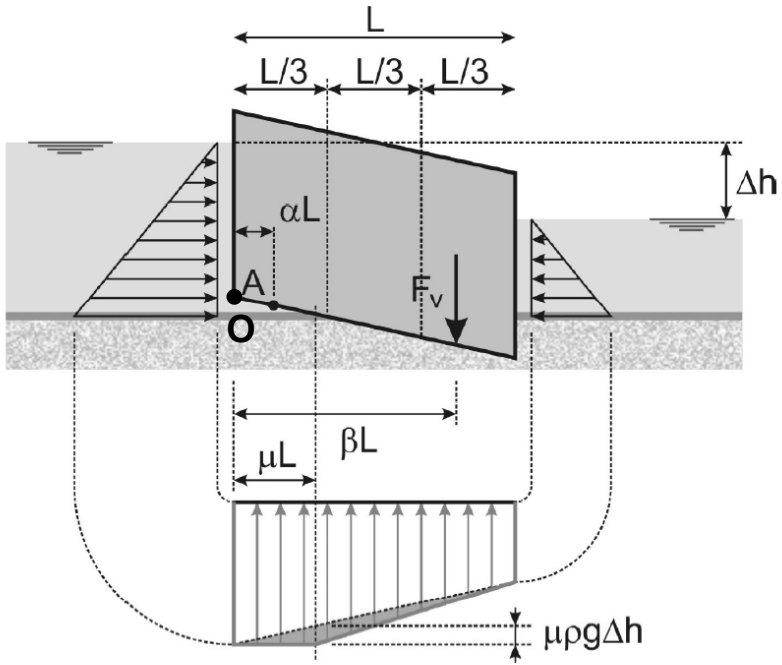
$$M_G = G \cdot \frac{14\text{m}}{6} = 7210\ \text{kN/m}$$

$$e = \frac{\Sigma M}{\Sigma F_v} = \frac{-3546\ \text{kNm/m}}{1922\ \text{kN/m}} = -1{,}8 < \frac{14\text{m}}{6}$$

No tension stresses at the foundation,
Tilting is verified

Kantelen / *Tilting*

Base becomes detached from the subsurface over a length *μL*.



Berekening μ en korrelspanning / *Calculation of μ and grain stress*

Stel $\alpha = 0$ / let $\alpha = 0$ → Point A=O
Extra opwaartse kracht door toename van de opwaartse waterdruk / *Additional upward force due to increase in upward water pressure*

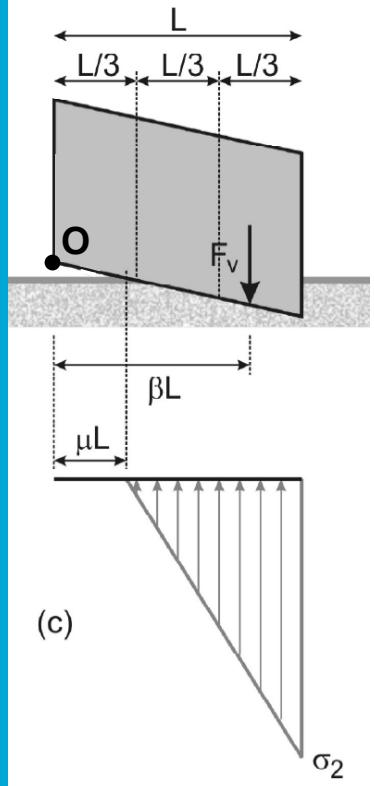
$$\Delta p = \mu \rho g \Delta h$$

Extra pressure = extra *F* and *M*

$$F_{\text{extra}} = \frac{1}{2}\mu B L \rho g \Delta h$$

$$M_{\text{extra}} = \mu(1+\mu)\frac{1}{6}BL^2\rho g \Delta h$$

Kantelen / *Tilting*



New equilibrium:

$$F_{\mathrm{v}} + F_{\mathrm{extra}} + F_{\mathrm{k}} = 0$$

$$M_{\mathrm{h}} + M_{\mathrm{v}}(\mathrm{O}) + M_{\mathrm{extra}} + M_k = 0$$

met F_v en M_v(O) exclusief de toename van de opwaartse waterdruk
with F_v and M_v(O) excluding the increase in upward water pressure

F_k and M_k are the force and moment associated with the soil (grain) stresses

Korrelspanning neemt lineair toe van nul tot σ_2 over het rechter $(1-\mu)L$ deel van het grondvlak van de constructie (in dit voorbeeld) / *Grain stress increases linearly from zero to σ_2 over the right $(1-\mu)L$ part of the base of the structure (in this example).*

Force: $$F_k = \frac{1}{2}\sigma_2(1-\mu)LB$$

Moment: $$M_k = (1-\mu)(2+\mu)\frac{1}{6}BL^2\sigma_2$$

Kantelen / *Tilting*

Berekening μ en korrelspanning / *Calculation of μ and grain stress*

$$F_v + F_{extra} + F_k = 0$$

$$M_h + M_v(0) + M_{extra} + M_k = 0$$

Ofwel / or

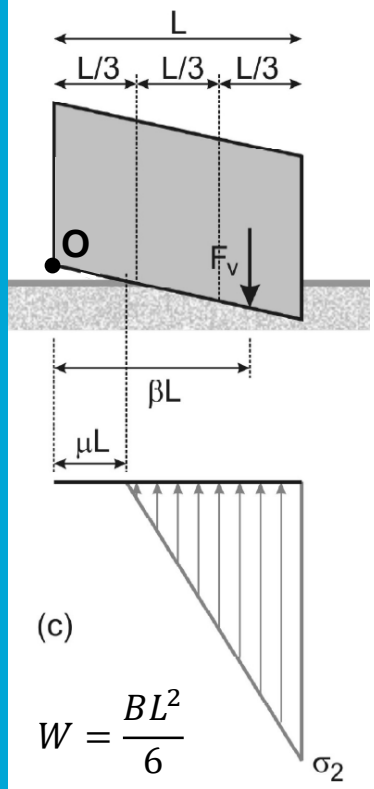
$$F_v + \frac{1}{2}\mu BL\rho g\Delta h + \frac{1}{2}\sigma_2(1-\mu)LB = 0$$

$$M_h + M_v(0) + \mu(1+\mu)\frac{1}{6}BL^2\rho g\Delta h + (1-\mu)(2+\mu)\frac{1}{6}BL^2\sigma_2 = 0$$

Solve for μ and σ_2 (with $W = BL^2/6$)

$$\mu = \frac{M_h + M_v(0) - \frac{2}{3}LF_v}{\frac{1}{3}LF_v + W\rho g\Delta h}$$

$$\sigma_2 = \frac{1}{1-\mu}\left(2\frac{F_v}{BL} + \rho g\mu\Delta h\right)$$



Kantelen / *Tilting* [Reflections]

- As the eccentricity increases, the max (grain) stress at the base (e.g. σ_2) rises.
- Once tilting initiates, the support reaction from the foundation becomes concentrated over a (progressively) smaller portion of the base.
- This concentration leads to
  - higher upward water pressure
  - higher risk of shear failure in the subsoil (which must be verified)
- If tilting results in downstream detachment, the uplift water pressure decreases, which is beneficial for stability.

The vertical effective soil stress, required to resist the acting loads (σ_{max}), should not exceed the maximum bearing capacity of the soil (p'_{max}), otherwise the soil will collapse:

$$\sigma_{max} < p'_{max}$$

The bearing capacity of densely packed sand is often assumed to be 500 kN/m^2, with design value no larger than 400 kN/m^2 (CUR, 2010).

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Questions?



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Paalfundering / *Pile foundation*

1. Aannamen / *Assumptions*

- Constructie is oneindig stijf / *construction is infinitely rigid*
- Palen gedragen zich als identieke lineaire veren / *piles behave like linear springs*

2. Uitgangspunten / *Basic principles*

- Palen staan in een patroon van regelmatige rechthoeken / *The piles are arranged in a pattern of regular rectangles*.
- *n* palen in de doorsnede en m palen daar loodrecht op / *n piles in the cross-section and m piles perpendicular to them*
- Momenten t.o.v midden van het grondvlak van de constructive (α = 1/2) / *moments with respect to the center of the base of the structure (α = 1/2)*
- Afschuiven blijft buiten beschouwing / *Sliding remains out of consideration*






Paalfundering / *Pile foundation*

Evenwicht / *Equilibrium*

Vertikaal / *Vertical*

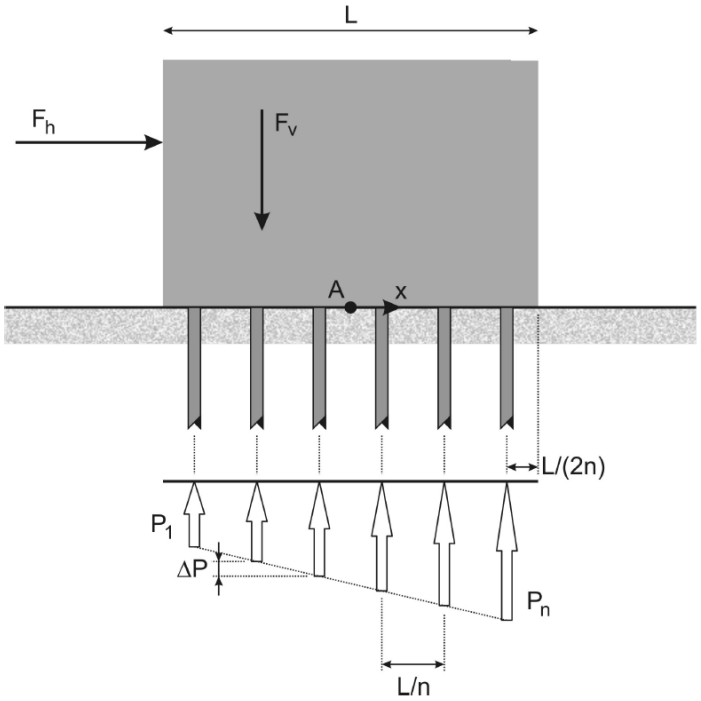
$$\frac{F_v}{m} + \sum_{i=i}^{n} P_i = 0$$

Momenten / *Moments*

$$\frac{M_v(1/2) + M_h}{m} + \sum_{i=i}^{n} x_i P_i = 0$$

met / *with*

$$x_i = \left(i - \frac{n+1}{2}\right)\frac{L}{n}$$



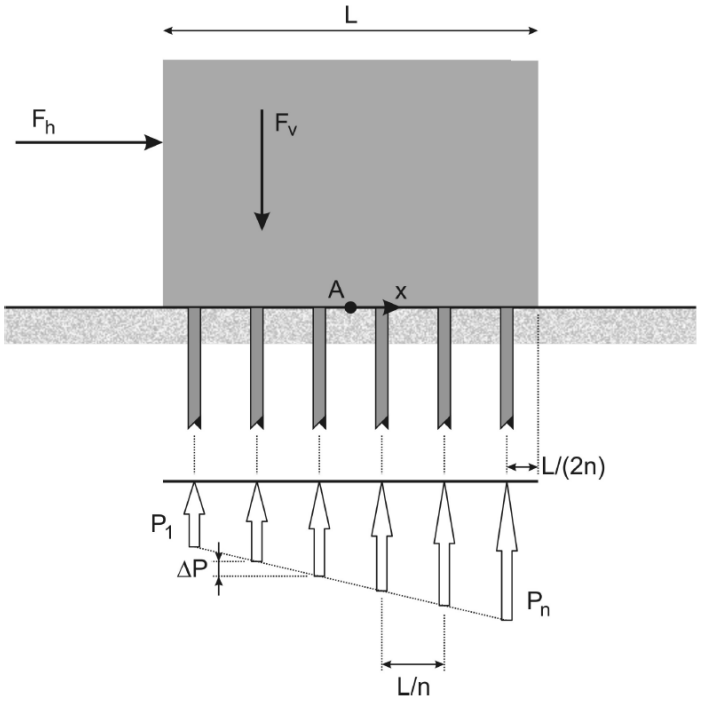
Paalfundering / *Pile foundation*

Average for on each pile:

$$P_{\text{gem}} = \frac{F_{\text{v}}}{m \cdot n}$$

$$\Delta P = \frac{12}{n^2 - 1} \frac{M_{\text{h}} + M_{\text{v}}(1/2)}{m \cdot L}$$

$$P_i = \frac{1}{m}\left[-\frac{F_{\text{v}}}{n} - \left(i - \frac{n+1}{2}\right)\Delta P\right]$$

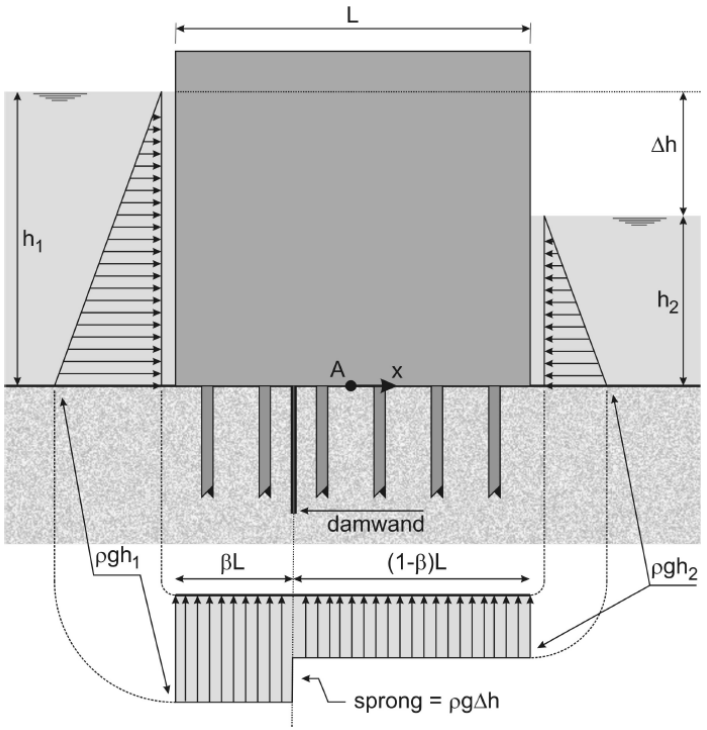


Full derivation of these equations in textbook (Section 8.5)

Paalfundering / *Pile foundation*

(Virtually) no contact between the base of the structure and the soil

1. **Piping** can be a problem!!! → Seepage screen to prevent this flow
2. The flow intensity decreases with increasing length of the barrier.
3. **No friction**: all horizontal loads must be absorbed by the piles
4. Because of no contact between soils and structure, gradient of under-pressures ~ 0



Uplift force:

$$F_{op} = \rho g B L [h_1 \beta + h_2 (1 - \beta)]$$

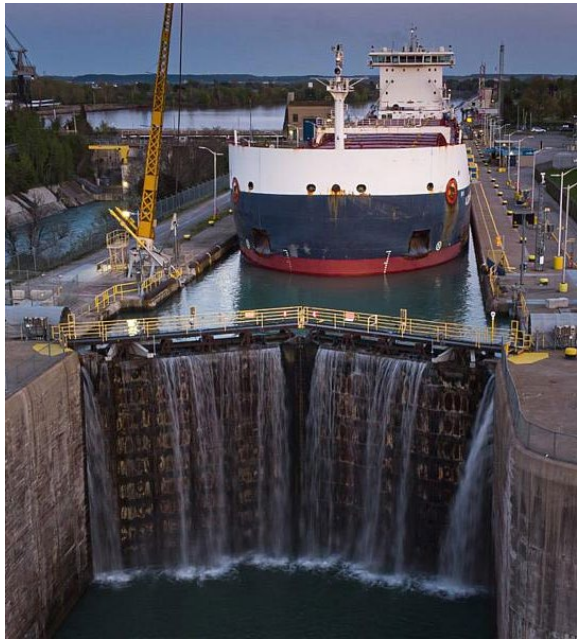
Questions?

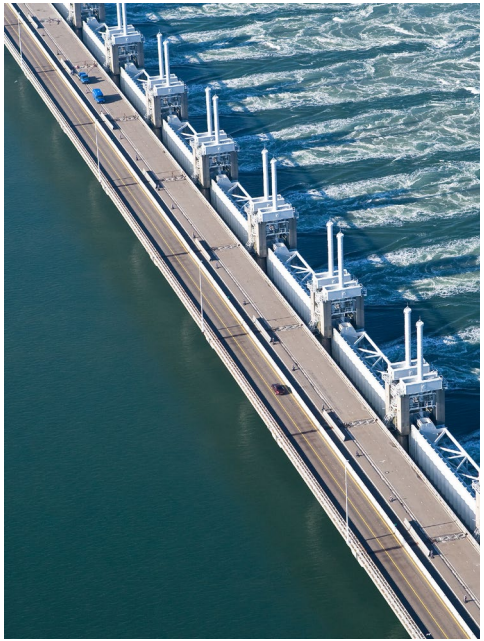


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Water in de constructie / *Water in the structure*







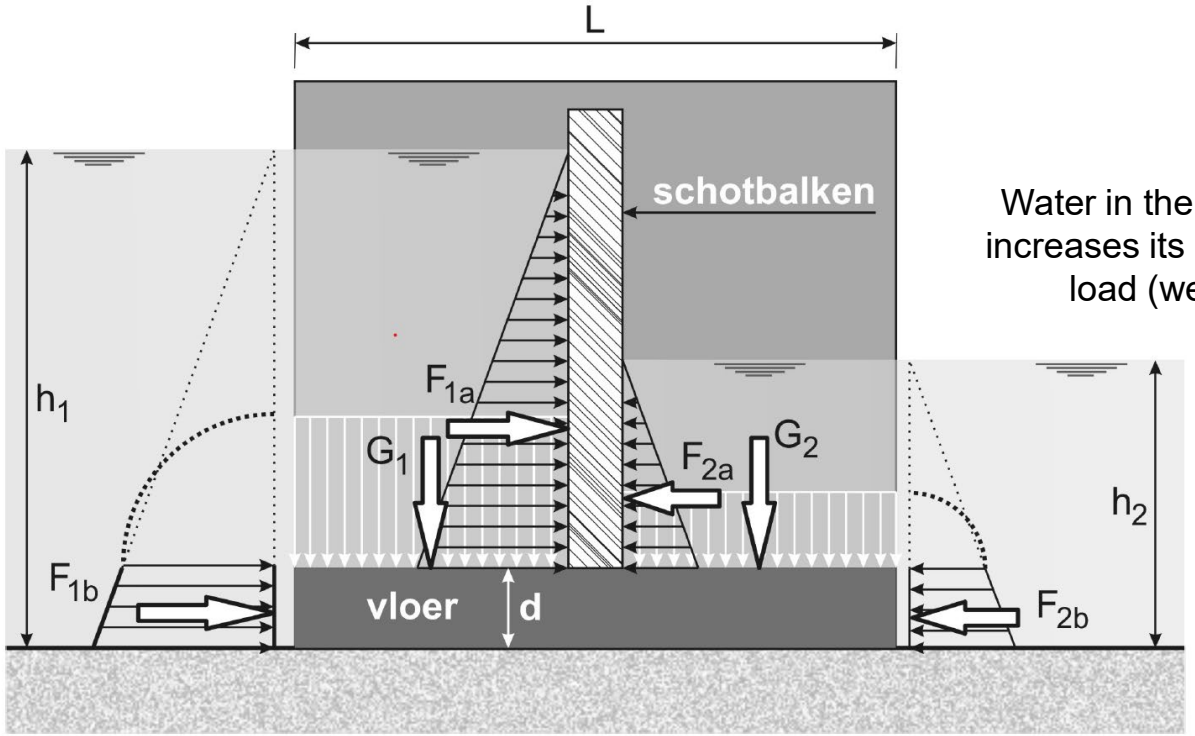


Hydraulic Structures are meant to get wet!!

Water through the structure affects the loads

Water in de constructie / *Water in the structure*

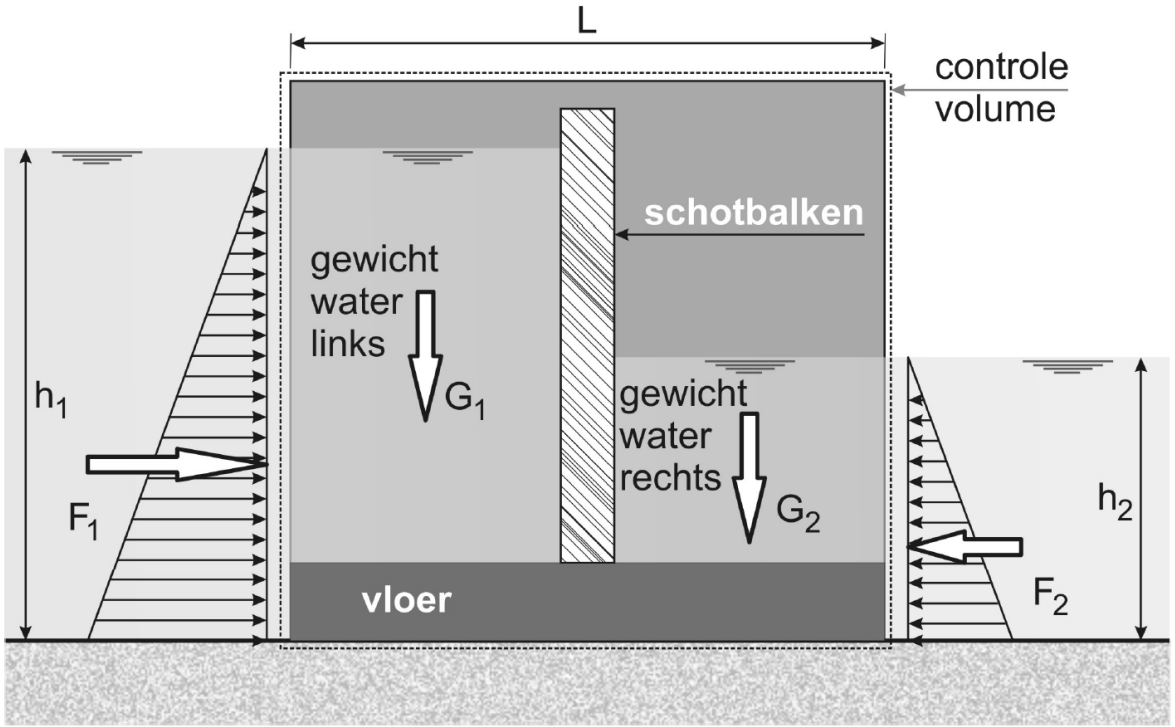
Waterdruk / *Water pressure*



Water in the structure increases its downward load (weight)

Water in de constructie / *Water in the structure*

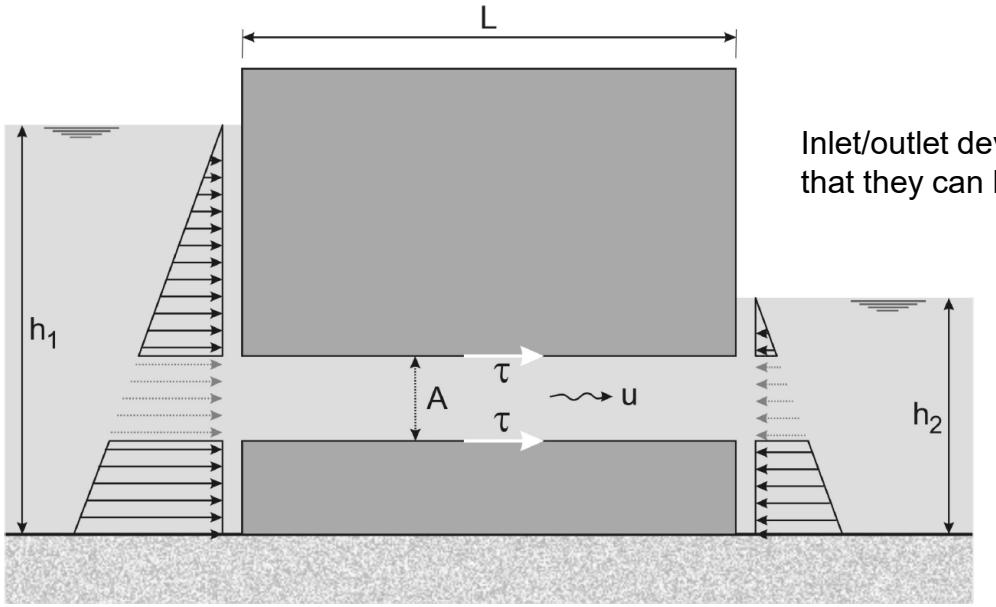
Waterdruk / *Water pressure*



Additional loads (not shown): (1) **Wave loads**; (2) **Under-pressures** (uplift)

Stroming door de constructie / *Flow through the structure*

Druk en wandwrijving / *Pressure and wall friction* (Steady flow)



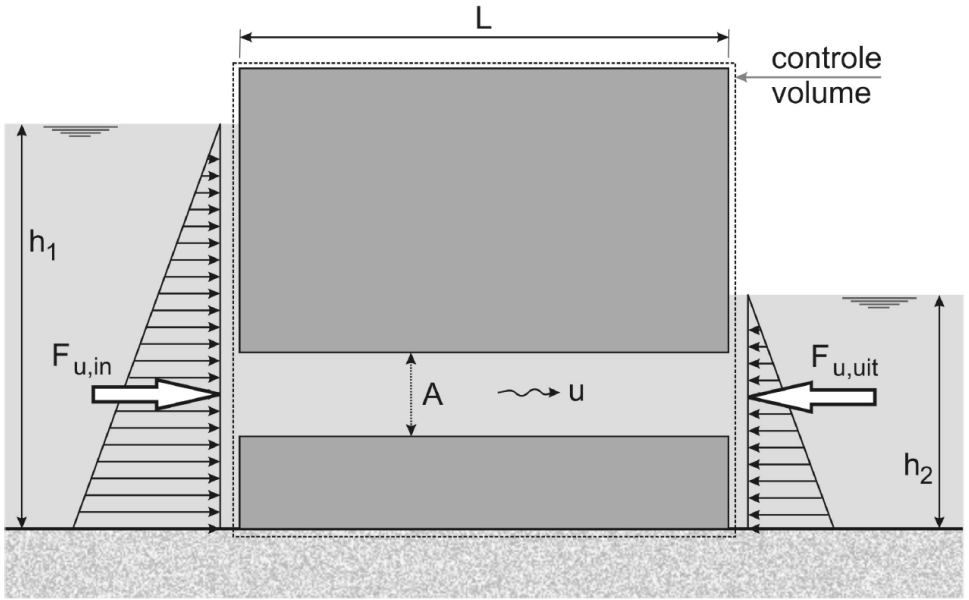
Inlet/outlet deviations are so small that they can be neglected.

- Hydrostatic load for completely closed structure
- Subtract the (missing) loads due to the hole in the structure (in this case F_b),
- Add the frictional force exerted by the flow on the wall of the pipe (in this case equal to F_b)

Stroming door de constructie / *Flow through the structure*

Gebruik controle volume / *Use control volume*

$$F_{\text{res}} = \frac{1}{2}\rho g h_1^2 B + F_{\text{u,in}} - \frac{1}{2}\rho g h_2^2 B - F_{\text{u,out}}$$



Conservation of momentum

$$F_{\text{u,in}} = \rho u^2 A$$

$$F_{\text{u,out}} = -\rho u^2 A$$

$$\downarrow$$

$$F_{\text{res}} = \frac{1}{2}\rho g h_1^2 B - \frac{1}{2}\rho g h_2^2 B$$

(same as impervious structure)

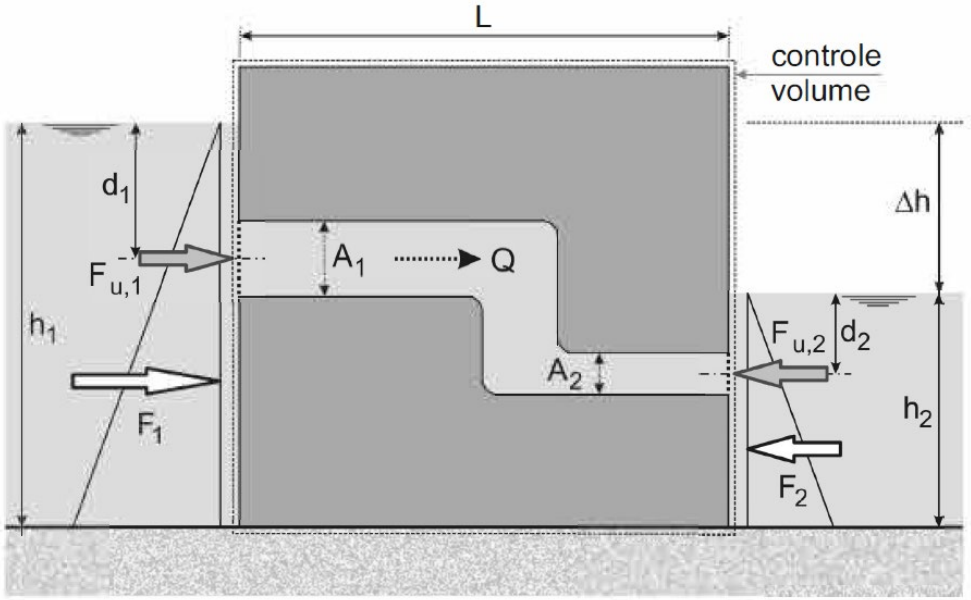
Attention: (1) Self-weight of the structure
(2) There is (still) a torque that contributes to M_h

Stroming door de constructie / *Flow through the structure*

Asymmetrie / *Asymmetry*

high-water side, water flows into the control volume at an average velocity Q/A_1, and on the low-water side, it exits at velocity Q/A_2.

$$F_{\text{res}} = \frac{1}{2}\rho g h_1^2 B + \rho\frac{Q^2}{A_1} - \frac{1}{2}\rho g h_2^2 B - \rho\frac{Q^2}{A_2}$$



Continuity Eq.
$Q = u_1A_1 = u_2A_2$

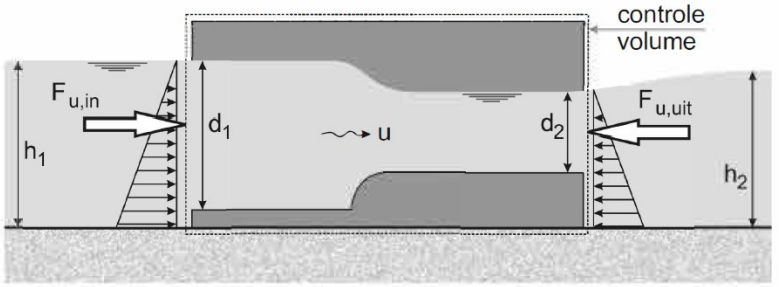
$$F_{u,1} = \rho u_1^2 A_1 = \rho\frac{Q^2}{A_1}$$

$$F_{u,2} = \rho u_2^2 A_2 = \rho\frac{Q^2}{A_2}$$

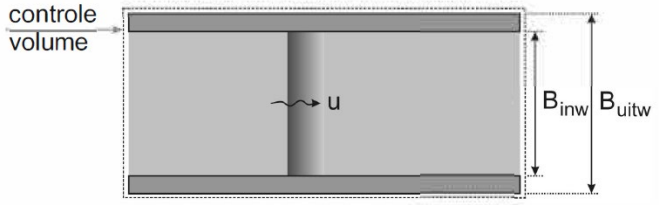
Stroming door de constructie / *Flow through the structure*

Voorbeeld: overlaat / *Example: spillway (open channel flow)*

$$F_{\text{res}} = \frac{1}{2}\rho g h_1^2 B_{\text{uitw}} + \rho \frac{Q^2}{h_1 B_{\text{inw}}} - \frac{1}{2}\rho g h_2^2 B_{\text{uitw}} - \rho \frac{Q^2}{h_2 B_{\text{inw}}}$$



langsdoorsnede



bovenaanzicht

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Questions?



CTB2410 Waterbouwkunde

(Week 2)

Dr. ir. Davide Wüthrich

Assistant Professor in Hydraulic Structures

TU Delft



